Credit Risk Monitor - Solution Overview

Live app: **credit-risk-monitor-romit.streamlit.app** | Repo: **github.com/romitsoni/credit-risk-dashboard** | Synthetic portfolio: 10,000 customers, 14,685 loans, 601.1M exposure, 16.1% default rate

1. What the dashboard shows and how an analyst uses it

Sidebar filters (loan type, region, employment, income band, risk segment) apply to every tab, including AI answers.

- **Portfolio** - Where is the risk? KPIs, exposure and default rate by risk segment, profile by any dimension, cross-cut heatmap, origination-year trends, drill-down to loans, 5 insights. *Use:* find the riskiest group, then drill into its loans.
- **Account** - Why is this loan risky? PD from both models, SHAP drivers with the customer's real values, 12-month repayment chart, bureau lines, other loans, flags. *Use:* explain a score before acting.
- **Ask the Portfolio** - plain-English questions, answered with exact figures and the source table.
- **Anomaly Review** - daily review list of unusual accounts: reason, Confirm / Dismiss, decision log.
- **Collections** (extra) - which actions recover money, with confidence bars and a cost calculator.
- **Model & Monitoring** - metrics, lift, calibration, drivers, PSI drift check, monitoring plan, fairness.

2. AI feature 1 - Ask the Portfolio (LLM)

How: Gemini call 1 sees only the question + a list of approved tools (no data) and returns a tool or "decline". Pandas computes the numbers with the same functions as the charts. Call 2 sees only that small aggregated table and writes 1-3 sentences. Code checks **every number** in the answer against the table; any mismatch -> a code-written answer. Shown with a badge (Verified / Safe answer / Declined) and the source. No key or API down -> quick questions, no crash.

Trigger	False positive	False negative	Human step
Analyst types a question	Answers with the wrong tool (city tier vs region); source box exposes it	Declines an answerable question; rephrase or use quick questions	Check badge + source table before using a number

3. AI feature 2 - Anomaly detection

How: each loan vs its **own history** (clean months 1-9 then a miss in 10-12; DPD up 30+ days) and its **peer segment** (top 1% bureau utilization or bureau DPD within loan type x income band), plus Isolation Forest for the rarest 2%. Every flag has a plain-English reason. 981 loans flagged; they default at **23.3% vs 15.6%** for the rest.

Trigger	False positive	False negative	Human step
Any rule above	Flagged but fine, e.g. one late month from a salary delay	Slow, steady decline that never crosses a threshold	Read reason, Confirm / Dismiss + note; logged with who and when

4. Model results and approach

Model (test set, split by customer)	AUC	KS	Gini	5-fold CV AUC
Logistic regression (baseline)	0.6625	0.2563	0.3249	0.672
XGBoost (dashboard; SHAP drivers)	0.6611	0.2434	0.3223	0.676

- **Data:** repayment (12 rows/loan) and bureau (n lines/customer) summarized first, then left-joined to loans -> 1 row per loan, row count asserted. Group-median fills + missing flags; nothing dropped.
- **Features:** lateness, missed months, DPD trend, pay ratio; bureau write-offs, DPD, utilization; income, age, employment, loan-to-income.
- **Excluded:** collections (bank's reaction to non-payment = leakage), gender and marital status (fair lending; AUC did not drop), unreliable dates.
- **Quality:** top 20% of scores hold 36.7% of defaults; predicted 15.7% vs actual 16.1%; segments rank 8% / 15% / 25% / 39%.
- **Why AUC is moderate:** both models hit the same ceiling; Kaggle Home Credit feature ideas added +0.002 (noise); missed defaults look like good customers in every table. The limit is the data.

5. Top 3 portfolio insights

- **Mortgages are the riskiest and biggest book:** 24.7% default vs 16.1% overall, and 52% of exposure. *So what:* review mortgage underwriting first.
- **Income and age drive risk steadily:** lowest income band 24.6% vs highest 9.7%; under-30s 21.6% vs 60+ 6.8%; region barely matters (15.0-16.9%). Cross-cut: low-income contract workers 27.6% but 12.0% one band up. *So what:* loan-to-income limits and stricter income checks for the lowest band; rules by profile, not geography.
- **The bank does not price for risk:** interest rate alone has AUC 0.507; good and bad customers pay almost the same rate within each loan type. *So what:* use the score for risk-based pricing.

6. Monitoring after deployment

- Monthly PSI on the score and key inputs (>0.10 watch, >0.25 retrain); monthly calibration and segment ranking; quarterly AUC/KS on matured loans (alert if AUC drops >0.03); data-quality checks on every load; quarterly fairness check. A human approves every retrain.

7. Key assumptions

- The model is a **behaviour score** for existing loans with 12 months of history, not an application score.
- **target_default** was observed after the 12 repayment months.
- Dates do not line up across tables (45% of collection events precede the loan; 62% of bureau lines open after it), so no date features are used; the bureau file is treated as a current snapshot.
- Exposure = original loan amount; income is annual.
- Collection outcome labels are unreliable (75.7% of "Payment Received" recovered nothing), so collections are analysed separately and "paid" = amount recovered > 0.
- Anomaly decisions are stored in a CSV for the demo (resets on cloud restart).

8. What we need from the bank's IT and data teams

- **Data:** a governed daily feed of loans, repayments and bureau data with reliable, consistent dates; default dates, not only a 0/1 flag; 24+ months of history; bureau data as at approval date; collection outcomes captured from the payment system.
- **Extra signals:** bank-statement income and cash-flow data, current balances for exposure, and costs per collection action.
- **Platform:** hosting inside the bank network, single sign-on and role-based access, a database for analyst decisions with an audit trail, scheduled scoring and monitoring jobs, secrets management.
- **LLM:** an enterprise or private LLM deployment (no training on bank data), logging of every question and answer, and a security review.
- **Governance:** independent model validation, a named model owner, fair-lending review, and an approval process for retraining and threshold changes.

9. Collections strategy (extra)

- All actions recover money in 23-27% of cases (only Settlement vs SMS is a real gap), and the default score does not predict who pays (AUC 0.49). Recommendation: least-cost escalation, a separate collections score, better outcome capture, and champion / challenger tests.